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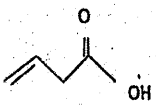
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COMMON QUARTERLY EXAMINATION - 2025

Standard - 12 Reg.No. | | | | | | |

Time Allowed:3.00 Hours **CHEMISTRY** *Maximum Marks:70*

I. Choose the correct answer: **15×1=15**

1) Zinc is obtained from ZnO by
   a) Carbon reduction b) Reduction using Silver
   c) Electrochemical process d) Acid leaching
2) Duralumin is an alloy of
   a) Cu, Mn b) Cu, Al, Mg
   c) Al, Mn d) Al, Cu, Mn, Mg
3) Assertion : Bond dissociation energy of fluorine is greater than chlorine gas.
   Reason : Chlorine has more electronic repulsion than fluorine.
   a) both assertion and reason true and reason is the correct explanation of assertion.
   b) both assertion and reason are true but reason is not the correct explanation of assertion.
   c) assertion is true but reason is false.
   d) both assertion and reason are false.
4) Hybridisation of $XeOF_2$ is
   a) Sp^3d^3 b) Sp^3d c) Sp^3d^2 d) Sp^3
5) The magnetic moment of Mn^{2+} ion is
   a) 5.92 BM b) 2.80 BM c) 8.95 BM d) 3.90 BM
6) The yellow colour in NaCl crystal is due to
   a) excitation of electrons in F centres
   b) reflection of light from Cl^- ions on the surface
   c) refraction of light of from Na^+ ion
   d) all of the above
7) Isotropy nature exist in _____.
   a) crystalline solid b) amorphous solid c) both a and b d) none of these
8) After 2 hours a radioactive substance becomes $\left(\frac{1}{16}\right)^{th}$ of original around then the half life (in min) is
   a) 60 minutes b) 120 minutes c) 30 minutes d) 15 minutes
9) Surface area of reactant increases. The rate of reaction will be
   a) increases b) decreases c) constant d) none of these
10) The PH of 10^{-5} M KOH solution will be
   a) 9 b) 5 c) 19 d) none of these
11) In the reaction Ethanol $\xrightarrow{PCl_5}$ x $\xrightarrow{\text{alc KOH}}$ y $\xrightarrow{H_2SO_4/H_2O}$ z the 'z' is
   a) ethane b) ethoxy ethane c) ethyl bi sulphite d) ethanol
12) Which of the following compound can be used surgical anaesthetic agent in surgery?
   a) Diethyl ether b) Anisole c) Phenol d) Methanol
13) The formation of cyanohydrins from acetone is an example of
   a) Nueleophilic substitution b) Electrophilic substitution
   c) Electrophilic addition d) Nucleophilic addition
14) The IUPAC name of 
   a) but-3-enoic acid b) but-1-ene-4-oic acid
   c) but-2-ene-1-oic acid d) but-3-ene-1-oic acid

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15) Benzoic acid does not undergo friedel craft's reaction because
a) strong activating nature of carboxylic group
b) strong deactivating nature of carboxylic group
c) both a and b
d) none of these

II. Answer any six questions. Question No.24 is compulsory: **6×2=12**

16) What are the various steps involved in the extraction of pure metals from their ores?
17) Give the uses of sulphuric acid.
18) What is Inert pair effect?
19) Calculate the number of atoms in BCC unit cell.
20) Give two examples for Zero order reaction.
21) Write the preparation of acrolein.
22) How is phenol prepared from Chloro Benzene?
23) Write a note on Claisen Condensation.
24) Calculate PH of 10^{-7}M H_2SO_4.

III. Answer any six questions. Question No.33 is compulsory: **6×3=18**

25) Write a note on fisher tropsch synthesis.
26) Give the oxidation state of halogen in the following:
i) OF_2 ii) O_2F_2 iii) Cl_2O_3
27) What are Interstitial compounds?
28) Explain Schottky defect.
29) Write Arrhenius equation and explains the terms involved.
30) Explain common ion effect with an example.
31) Write a note on Coupling reaction.
32) Explain reducing action of formic acid.
33) Identify A, B, C.

$$CH_3COOH \xrightarrow{SOCl_2} A \xrightarrow{Pd/BaSO_4} B \xrightarrow{NaOH} C$$

IV. Answer all the questions: **5×5=25**

34) a) Explain Electrometallurgy of Aluminium. **(OR)**
b) i) What is Catenation? Describe briefly the catenation property of carbon.
ii) What happen when PCl_5 is heated?
35) a) Compare lanthanoids and actinolds. **(OR)**
b) i) Give the uses of Helium.
ii) Complete the following reactions:
a) $B_2H_6 + CH_3OH \longrightarrow$ b) $HCOOH + H_2SO_4 \longrightarrow$
36) a) Write short note on metal excess and metal deficiency defect with an example. **(OR)**
b) Derive Integrated rate law for a first order reaction.
37) a) i) Define ionic product of water. Give its value at room temperature.
ii) Write the expression for the solubility product of $Ca_3(PO_4)_2$. **(OR)**
b) How will you distinguish between 1°, 2° and 3° alcohols by Victor Mayer's Test.
38) a) i) Write a note on Schotten-Baumann reaction.
ii) Write the preparation of Picric acid from phenol. **(OR)**
b) i) Write the Mechanism of Cannizaro reaction.
ii) Write the test for Carboxylic acid group.

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